Name: _____

TOPIC TEST 1

Chapter 2 Discrete random variables

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Eight multiple-choice questions (8 marks).
- Part B: Eight short-answer questions (43 marks).
- Part C: Three analysis questions (19 marks).

Part A

Multiple-choice questions

Eight multiple-choice questions
1 mark each: 8 marks
Circle the correct answer.

1 A pair of dice is rolled and $X =$ the maximum number. Which of the following ordered pairs would *not* make up the probability function for X?

A $\left(3, \frac{5}{36}\right)$
B $\left(5, \frac{8}{36}\right)$
C $\left(1, \frac{1}{36}\right)$
D $\left(2, \frac{3}{36}\right)$
E $\left(6, \frac{11}{36}\right)$

[1 mark]

2 Which of the following is *not* an example of a discrete variable?

A The number of stairs between floors in buildings
B The number of snooker balls sunk in a break
C The time taken for a red light to turn green
D The number of leaves on an apple tree in an orchard
E The number of goals scored in a soccer match

[1 mark]

© Cengage Learning Australia Pty Ltd 2014

3 Which of the following could represent a probability distribution?

A

x	$p(x)$
1	0.44
2	0.25
3	0.32
4	0.01

B

x	$p(x)$
10	$\frac{3}{8}$
20	$\frac{1}{6}$
30	$\frac{2}{5}$
40	$\frac{1}{4}$

C

x	$p(x)$
-1	0.2
0	0.2
1	0.3
2	0.4

D

x	$p(x)$
2	$\frac{1}{9}$
5	$\frac{1}{3}$
7	$\frac{1}{9}$
8	$\frac{4}{9}$

E

x	$p(x)$
3	0.2
8	0.5
9	-0.1
11	0.4

[1 mark]

4 Consider the probability function $p(x) = \frac{2x - 1}{36}$ for $x = 1, 2, 3, \ldots, 6$.

What is $p(2 < x \leq 5)$?

A $\frac{20}{36}$

B $\frac{7}{12}$

C $\frac{2}{3}$

D $\frac{11}{36}$

E $\frac{15}{36}$

[1 mark]

© Cengage Learning Australia Pty Ltd 2014

5 A discrete random variable X can take the values 0, 2, 4, 6 and 8 with probabilities 0.1, 0.15, 0.15, 0.25 and 0.35 respectively. What is the expected value of X?

A 5.2

B 5.35

C 4.7

D 10.75

E 6.1

[1 mark]

6 A driver takes the same route to a shopping centre each day, passing through two sets of traffic lights. The driver knows that that there is a 60% chance the first light will be red. If the first light is red, there is only a 20% chance the second light will be red. If the first light is green, there is 70% chance the second light will also be green. How many red lights can the driver expect on the way to the shopping centre?

A 1.00

B 1.08

C 0.76

D 0.84

E 2.00

[1 mark]

7 A discrete random variable X can take the values 5, 6, 7 and 8 with probabilities 0.4, 0.3, 0.2 and 0.1 respectively. What is the variance of X?

A 2

B 1.25

C 1

D 1.5

E 1.4

[1 mark]

© Cengage Learning Australia Pty Ltd 2014

8 For the situation described in question 7, what is $\text{Var}(4X - 3)$?

A 10.6

B 13

C 1

D 5

E 16

[1 mark]

Part B

Short-answer questions

Six short-answer questions
56 marks
Show your working where appropriate.

1 The numbers of people living in the households of a street were found to be as follows:

4 3 1 2 6 2 4 5 2 4 4 3 2 3 3 1 3

3 2 3 4 3 4 2 4 2 2 4 3 1 4 4 2 4

2 4 4 5 4 4

If the number of people living in a household is the variable, X:

a construct a probability distribution for X

b calculate $P(2 \leq X < 5)$

c calculate $E(X)$.

[7 marks]

2 A spinner is in the shape of a regular pentagon. It has five equally sized segments numbered 2, 2, 5, 7 and 7. Let X be the number on which the pointer rests after a spin.

a Construct a probability distribution for X.

b Calculated $E(X)$.

c If the value of each number on the spinner is increased by 3, what is the new expected value of X?

d If the value of each of the original numbers on the spinner is increased by a factor of 5, what is the new expected value of X?

[6 marks]

3 A ball is randomly selected from a container holding 10 identical balls, each numbered 1, 2, 3, …, 10. Three selections are made, with the ball being returned to the container after each selection. A success is 'selecting a 7'.

a What is the probability of only 1 success in three selections?

b What is the probability that the only success will be in the last selection?

[6 marks]

4 The probability function for a discrete random variable, X, is

$p(x) = \dfrac{kx}{x+1}$ for $x = 1, 2, 3, 4$ and 5.

a Construct a probability distribution for X.

© Cengage Learning Australia Pty Ltd 2014

b Use the probability distribution to calculate the value of k.

[4 marks]

5 A large chocolate wheel at a district fair has the numbers 1, 2, 3, …, 60 on its circumference. The chocolate wheel is equally likely to finish with its pointer on any number on its circumference.

a Describe the probability distribution for X, the number on which the pointer finishes after a spin of the chocolate wheel.

b Calculate $P(X = 32)$.

c Calculate $P(17 \leq X \leq 44)$.

[5 marks]

6 Calculate $E(X)$, $\text{Var}(X)$ and σ for the following distribution:

x	2	3	5	9
$P(X = x)$	0.3	0.4	0.1	0.2

[6 marks]

7 A lottery with 1000 tickets gives one prize of \$100, three prizes of \$50 and five prizes of \$25 each.

a Find the expected winnings of a ticket.

b If each ticket costs \$1, what is the expected value of the game?

[3 marks]

© Cengage Learning Australia Pty Ltd 2014

8 A basketball player shoots two consecutive free-throws. Each free-throw is worth 1 point and has a probability of success of 0.8. Let X = the number of points scored.

a Construct the probability distribution for X.

b How many points can the basketball player expect to score?

[6 marks]

Part C

Analysis questions

Three analysis questions
19 marks
Show your working where appropriate.

1 A random variable, X, has the probability distribution shown in the table below:

x	0	1	2	3
$P(X = x)$	0.15	0.25	a	b

If the expected value of X is 1.93 find the values of a and b.

[4 marks]

2 A random variable X has the probability distribution shown in the table below:

x	4	6	8	10
$P(X = x)$	k	$5k$	$7k$	$3k$

a Calculate the value of k.

b Find Var(X).

[6 marks]

3 A shop sells gift vouchers valued at \$10, \$20, \$40, \$100 or \$200. The value in dollars of a gift voucher sold may be regarded as a random variable, X, with the following distribution:

x	10	20	40	100	200
$P(X = x)$	0.20	0.40	0.22	0.11	0.07

a Find the mean and the standard deviation of X.

b What is the probability that the value of the next voucher sold is less than \$40?

The shop owner is considering whether to discontinue selling \$10 and \$20 vouchers. If she did this, a proportion (p) of customers who presently buy \$10 and \$20 vouchers would then buy a \$40 voucher. Other such customers would not buy a voucher. As a result the value, in dollars, of sales would be a random variable, Y, with the following distribution:

y	0	40	100	200
$P(Y = y)$	$0.6(1 - p)$	$0.22 + 0.6p$	0.11	0.07

c Find the mean of Y in terms of p.

d A survey suggests that the value of p would be between 0.4 and 0.6. Using this information, advise the shop owner on the likely effect on takings if she decides to discontinue selling \$10 and \$20 vouchers.

[9 marks]

[Total marks: 70]

© Cengage Learning Australia Pty Ltd 2014

Answers

Part A – Multiple-choice questions [1 mark each]

1 Probability distribution =

x	1	2	3	4	5	6
$p(x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

Ordered pairs for the function $= \left(1, \frac{1}{36}\right), \left(2, \frac{3}{36}\right), \left(3, \frac{5}{36}\right), \left(4, \frac{7}{36}\right), \left(5, \frac{9}{36}\right), \left(6, \frac{11}{36}\right)$

$\left(5, \frac{8}{36}\right)$ is not one of the ordered pairs listed.

$\therefore$ **B**

2 Examine each variable listed.

A The number of stairs between floors in buildings can be counted and cannot take on in-between values, so it is discrete.

B The number of snooker balls sunk in a break can be counted and cannot take on in-between values, so it is discrete.

C The time taken for a red light to turn green depends on the accuracy of the instrument used to measure time. It can take on in-between values. It is not discrete.

D The number of leaves on an apple tree in an orchard can be counted and cannot take on in-between values, so it is discrete.

E The number of goals scored in a soccer game can be counted and cannot take on in-between values, so it is discrete.

C **[1 mark]**

3 The probabilities in parts A, B. C and D are all valid values. One probability in part E is negative and so the distribution in part E can't be a probability distribution.

For a probability distribution $\Sigma p(x) = 1$.

A: $\Sigma p(x) = 1.02$

B: $\Sigma p(x) = 1\frac{23}{120}$

C: $\Sigma p(x) = 1.1$

D: $\Sigma p(x) = 1$

$\therefore$ **D**

© Cengage Learning Australia Pty Ltd 2014

4 Probability distribution =

x	$p(x)$
1	$\frac{1}{36}$
2	$\frac{3}{36}$
3	$\frac{5}{36}$
4	$\frac{7}{36}$
5	$\frac{9}{36}$
6	$\frac{11}{36}$

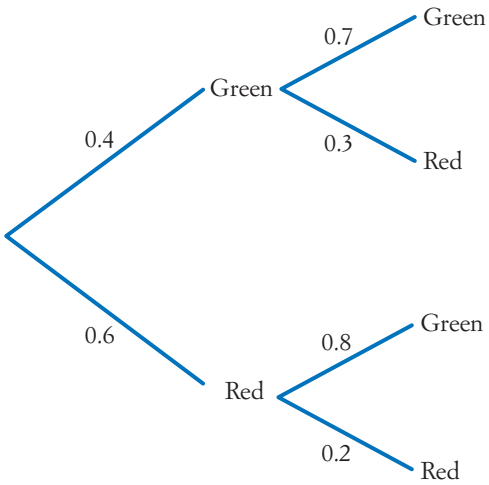
$$p(2 < x \leq 5) = p(3) + p(4) + p(5)$$
$$= \frac{5}{36} + \frac{7}{36} + \frac{9}{36}$$
$$= \frac{7}{12}$$

$\therefore$ **B**

5 $E(X) = 0 \times 0.1 + 2 \times 0.15 + 4 \times 0.15 + 6 \times 0.25 + 8 \times 0.35$
$= 5.2$

$\therefore$ **A**

6 Tree diagram for this situation =



$P(x = 0) = 0.4 \times 0.7 = 0.28$

$P(x = 1) = 0.6 \times 0.8 + 0.4 \times 0.3 = 0.6$

$P(x = 2) = 0.6 \times 0.2 = 0.12$

$E(X) = 0 \times 0.28 + 1 \times 0.6 + 2 \times 0.12$
$= 0.84$

$\therefore$ **D**

© Cengage Learning Australia Pty Ltd 2014

7 $E(X) = 5 \times 0.4 + 6 \times 0.3 + 7 \times 0.2 + 8 \times 0.1$

$= 6$

$E(X^2) = 5^2 \times 0.4 + 6^2 \times 0.3 + 7^2 \times 0.2 + 8^2 \times 0.1$

$= 37$

$\text{Var}(X) = 37 - 6^2 = 1$

$\therefore$ **C**

8 $\text{Var}(4X - 3) = 4^2 \times \text{Var}(X)$

$= 16 \times 1$

$= 16$

$\therefore$ **E**

Part B – Short-answer questions

1 a Construct a frequency table for the data. **[1 mark]**

x	1	2	3	4	5	6
f	3	10	9	15	2	1

$\Sigma f = 3 + 10 + 9 + 15 + 2 + 1 = 40$ **[1 mark]**

x	1	2	3	4	5	6
f	3	10	9	15	2	1
$p(x)$	$\frac{3}{40}$	$\frac{1}{4}$	$\frac{9}{40}$	$\frac{3}{8}$	$\frac{1}{20}$	$\frac{1}{40}$

[1 mark]

b $2 \leq x < 5$ means $x = 2, 3$ or 4 **[1 mark]**

$P(2 \leq x < 5) = P(x = 2) + P(x = 3) + P(x = 4)$

$= \frac{1}{4} + \frac{9}{40} + \frac{3}{8}$

$= \frac{17}{20}$ **[1 mark]**

c $E(X) = \Sigma x \,.\, p(x)$ **[1 mark]**

$= 1 \times \frac{3}{40} + 2 \times \frac{1}{4} + 3 \times \frac{9}{40} + 4 \times \frac{3}{8} + 5 \times \frac{1}{20} + 6 \times \frac{1}{40}$

$= 3\frac{3}{20}$ **[1 mark]**

2 a Construct the probability distribution.

x	2	5	7
$p(x)$	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{2}{5}$

[1 mark]

© Cengage Learning Australia Pty Ltd 2014

b $E(X) = 2 \times \frac{2}{5} + 5 \times \frac{1}{5} + 7 \times \frac{2}{5}$

$= 4\frac{3}{5}$ **[1 mark]**

c $E(X + b) = E(X) + b$

Let $Y = X + 3$

$E(Y) = E(X) + 3$ **[1 mark]**

$= 4\frac{3}{5} + 3$

$= 7\frac{3}{5}$ **[1 mark]**

d $E(bX) = bE(X)$

Let $Z = 5X$

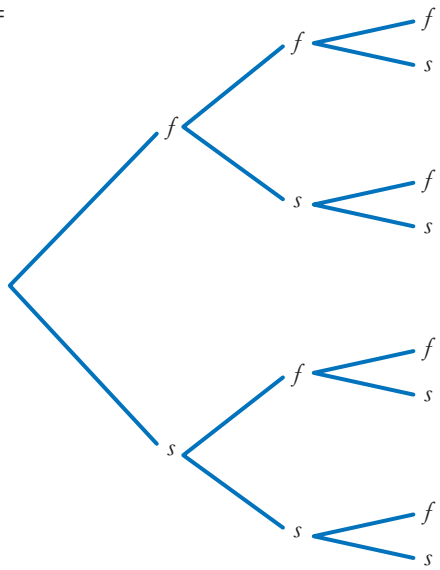
$E(Z) = 5E(X)$ **[1 mark]**

$= 4\frac{3}{5} \times 5$

$= \frac{23}{5} \times 5$

$= 23$ **[1 mark]**

3 a Tree diagram for this situation =



[1 mark]

Let $P(\text{success}) = p$ and $P(\text{failure}) = q$.

$p = \frac{1}{10}$ and $q = \frac{9}{10}$ **[1 mark]**

$P(\text{only 1 success}) = P(s, f, f) + P(f, s, f) + P(f, f, s)$ **[1 mark]**

$= p.q.q + q.p.q + q.q.p$

$= \frac{1}{10} \times \frac{9}{10} \times \frac{9}{10} + \frac{9}{10} \times \frac{1}{10} \times \frac{9}{10} + \frac{9}{10} \times \frac{9}{10} \times \frac{1}{10}$

$= 0.243$ **[1 mark]**

b $P(\text{only success in last draw}) = P(f, f, s)$ **[1 mark]**

$= q.q.p$

$= \frac{9}{10} \times \frac{9}{10} \times \frac{1}{10}$

$= 0.081$ **[1 mark]**

4 a Probability distribution =

x	1	2	3	4	5
$p(x)$	$\frac{k}{2}$	$\frac{2k}{3}$	$\frac{3k}{4}$	$\frac{4k}{5}$	$\frac{5k}{6}$

[1 mark]

b $\Sigma p(x) = 1$.

$\frac{k}{2} + \frac{2k}{3} + \frac{3k}{4} + \frac{4k}{5} + \frac{5k}{6} = 1$ **[1 mark]**

$\frac{60}{213} \times (\frac{k}{2} + \frac{2k}{3} + \frac{3k}{4} + \frac{4k}{5} + \frac{5k}{6}) = \frac{60}{213} \times 1$ **[1 mark]**

$\frac{30k + 40k + 45k + 48k + 50k}{60} = 1$

$\frac{213k}{60} = 1$

$k = \frac{60}{213}$

$k = \frac{20}{71}$ **[1 mark]**

5 a Each of the outcomes is equally likely.

The distribution is uniform. **[1 mark]**

b $n = b - a + 1$

$= 60 - 1 + 1$

$= 60$ **[1 mark]**

$P(X = x) = \frac{1}{n} = \frac{1}{60}$ **[1 mark]**

c $P(c \leq x \leq d) = \frac{d - c + 1}{b - a + 1}$ **[1 mark]**

$= \frac{44 - 17 + 1}{60} = \frac{7}{15}$ **[1 mark]**

© Cengage Learning Australia Pty Ltd 2014

6 $E(X) = \Sigma x \,.\, p(x)$

$= 2 \times 0.3 + 3 \times 0.4 + 5 \times 0.1 + 9 \times 0.2$

$= 4.1$ **[1 mark]**

$E(X^2) = 4 \times 0.3 + 9 \times 0.4 + 25 \times 0.1 + 81 \times 0.2$

$= 23.5$ **[1 mark]**

$\text{Var}(X) = E(X^2) - E(X)^2$ **[1 mark]**

$= 23.5 - 4.1^2$

$= 6.69$ **[1 mark]**

$\sigma = \sqrt{\text{Var}(X)}$ **[1 mark]**

$= \sqrt{6.69}$

≈ 2.59

$E(X) = 4.1$, $\text{Var}(X) = 6.69$ and $\sigma \approx 2.59$ **[1 mark]**

7 a Expected winnings $= E(X) = 100 \times \frac{1}{1000} + 50 \times \frac{3}{1000} + 25 \times \frac{5}{1000}$ **[1 mark]**

$= 37.5$

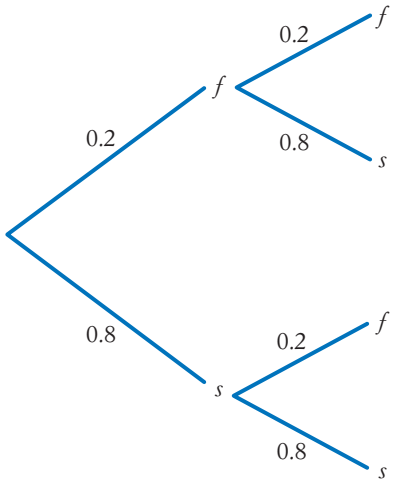
Expected winnings $= 37.5$ cents **[1 mark]**

b Expected value = Ticket price − expected winnings

$= \$1 - 37.5\text{c}$

$= -62.5$ cents **[1 mark]**

8 a Let s = success and f = failure.



[1 mark]

$P(X = 0) = P(f \text{ then } f) = 0.2 \times 0.2 = 0.04$

$P(X = 1) = P(s \text{ then } f) + P(f \text{ then } s)$

$= 0.8 \times 0.2 + 0.2 \times 0.8$

$= 0.32$

$P(X = 2) = P(s \text{ then } s) = 0.8 \times 0.8 = 0.64$ **[1 mark]**

x	0	1	2
$P(X = x)$	0.04	0.32	0.64

[1 mark]

b $E(X) = \Sigma x \,.\, p(x)$ **[1 mark]**

$= 0 \times 0.04 + 1 \times 0.32 + 2 \times 0.64$

$= 1.6$ **[1 mark]**

Each free throw is worth a point, so the basketball player can expect to score 1.6 points.

[1 mark]

Part C – Analysis questions

1 The sum of the probabilities must be 1.

$0.15 + 0.25 + a + b = 1$ **[1 mark]**

$a + b = 0.6$

$a = 0.6 - b$

$E(X) = (0 \times 0.15) + (1 \times 0.25) + 2a + 3b$

$1.93 = 0.25 + 2a + 3b$

$2a + 3b = 1.68$ **[1 mark]**

$2(0.6 - b) + 3b = 1.68$

$1.2 - 2b + 3b = 1.68$

$b = 0.48$ **[1 mark]**

$a = 0.6 - 0.48 = 0.12$ **[1 mark]**

2 a The sum of the probabilities must be 1.

$k + 5k + 7k + 3k = 1$ **[1 mark]**

$16k = 1$

$k = \frac{1}{16}$ **[1 mark]**

b Probability distribution =

x	4	6	8	10
$P(X = x)$	$\frac{1}{16}$	$\frac{5}{16}$	$\frac{7}{16}$	$\frac{3}{16}$

[1 mark]

© Cengage Learning Australia Pty Ltd 2014

$E(X) = 4 \times \frac{1}{16} + 6 \times \frac{5}{16} + 8 \times \frac{7}{16} + 10 \times \frac{3}{16}$

$= \frac{120}{16}$

$= \frac{15}{2}$

$= 7.5$ **[1 mark]**

$E(X^2) = 16 \times \frac{1}{16} + 36 \times \frac{5}{16} + 64 \times \frac{7}{16} + 100 \times \frac{3}{16}$

$= \frac{944}{16}$

$= 59$ **[1 mark]**

$\text{Var}(X) = 59 - 7.5^2$

$= 2.75$ **[1 mark]**

3 a $E(X) = (10 \times 0.20) + (20 \times 0.40) + (40 \times 0.22) + (100 \times 0.11) + (200 \times 0.07)$

$= 43.80$ **[1 mark]**

$E(X^2) = (10^2 \times 0.20) + (20^2 \times 0.40) + (40^2 \times 0.22) + (100^2 \times 0.11) + (200^2 \times 0.07)$

$= 4432$ **[1 mark]**

$\text{Var}(X) = 4432 - 43.8^2$

$= 2513.56$ **[1 mark]**

$\sigma \approx \sqrt{2513.56} \approx 50.14$

The mean value of a gift voucher is \$43.80 and the standard deviation is about \$50.14. **[1 mark]**

b There are two vouchers worth less than \$40.

$P(X < 40) = P(X = 1 \text{ or } 2)$

$= 0.20 + 0.40$

$= 0.6$ **[1 mark]**

c $E(Y) = [0 \times 0.6(1 - p)] + [40 \times (0.22 + 0.6p)] + [100 \times 0.11] + [200 \times 0.07]$ **[1 mark]**

$= 8.8 + 24p + 11 + 14$

$= 33.8 + 24p$ **[1 mark]**

The maximum value of p is 0.6.

The maximum value of $E(Y) = 33.8 + 24 \times 0.6$

$= 48.2$ **[1 mark]**

Her takings are likely to increase from \$43.80 to \$48.20. **[1 mark]**

© Cengage Learning Australia Pty Ltd 2014